Precalculus Review for Calculus

ANSWER KEY

Section 1: Functions, Domain, and Composition

1 Given $f(x) = \sqrt{x-3}$ and $g(x) = x^2 - 1$, find the domain of $(f \circ g)(x)$.

a. $(-\infty, -2] \cup [2, \infty)$ ✓

b. $[2, \infty)$

c. $[3, \infty)$

d. $[-2, 2]$

$(f \circ g)(x) = \sqrt{g(x) - 3} = \sqrt{x^2 - 4}$. For real output, $x^2 - 4 \geq 0 \Rightarrow |x| \geq 2$, so $x \in (-\infty, -2] \cup [2, \infty)$.

2 If $f(x) = \frac{2x+1}{x-3}$, what is $f^{-1}(x)$?

a. $f^{-1}(x) = \frac{3x+1}{x-2}$ ✓

b. $f^{-1}(x) = \frac{x-3}{2x+1}$

c. $f^{-1}(x) = \frac{3x-1}{x+2}$

d. $f^{-1}(x) = \frac{2x-1}{x+3}$

$y = \frac{2x+1}{x-3} \Rightarrow yx - 3y = 2x + 1 \Rightarrow x(y-2) = 3y + 1 \Rightarrow x = \frac{3y+1}{y-2}$. Thus $f^{-1}(x) = \frac{3x+1}{x-2}$.

3 Simplify the difference quotient $\frac{f(x+h) - f(x)}{h}$ for $f(x) = 3x^2 - 5x$.

a. $6x + 3h - 5$ ✓

b. $6x - 5$

c. $3x + 3h - 5$

d. $6x + h - 5$

$\frac{3(x+h)^2 - 5(x+h) - (3x^2 - 5x)}{h} = \frac{6xh + 3h^2 - 5h}{h} = 6x + 3h - 5$.

Section 2: Exponentials, Logarithms, and Trig Review

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4 Solve the equation $4^{x+1} = 8^{2x-1}$ for x.

a. $5/4$ ✓

b. 1

c. $3/2$

d. $2/3$

Write in base 2: $2^{2(x+1)} = 2^{3(2x-1)} \Rightarrow 2x+2 = 6x-3 \Rightarrow 4x = 5 \Rightarrow x = 5/4$.

5 Solve $\ln(2x-1) + \ln(x) = \ln(3)$.

a. $x = 3/2$ ✓

b. $x = 1$

c. $x = 2$

d. $x = 3$

$\ln(2x^2 - x) = \ln(3) \Rightarrow 2x^2 - x - 3 = 0 \Rightarrow (2x-3)(x+1) = 0$. Since $x > 1/2$, the only valid solution is $x = 3/2$.

6 Find the exact value of $\sin\left(\frac{7\pi}{6}\right) + \cos\left(\frac{4\pi}{3}\right)$.

a. -1 ✓

b. $-1/2$

c. 0

d. $-\sqrt{3}$

$\sin(7\pi/6) = -1/2$ and $\cos(4\pi/3) = -1/2$. Sum $= -1/2 + (-1/2) = -1$.

Section 3: Rational Functions, Holes, and Asymptotes

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7 Identify the vertical and horizontal asymptotes of $f(x) = \frac{3x^2 - 12}{x^2 - 5x + 6}$.

a. **Vertical:** $x = 3$**; Horizontal:** $y = 3$ ✓

b. Vertical: $x = 2, x = 3$; Horizontal: $y = 3$

c. Vertical: $x = 3$; Horizontal: $y = 0$

d. Vertical: $x = 2$; Horizontal: $y = 1$

Factor: $\frac{3(x-2)(x+2)}{(x-2)(x-3)} = \frac{3(x+2)}{x-3}$ for $x \neq 2$. Removable hole at $x = 2$, vertical asymptote at $x = 3$. As $x \to \infty$, $y \to 3$.

8 Determine the slant asymptote of $g(x) = \frac{2x^2 + 5x - 1}{x + 1}$.

a. $y = 2x + 3$ ✓

b. $y = 2x + 5$

c. $y = 2x - 1$

d. $y = x + 3$

Polynomial long division: $\frac{2x^2 + 5x - 1}{x + 1} = 2x + 3 - \frac{4}{x + 1}$. Slant asymptote is $y = 2x + 3$.

9 Find the domain of $h(x) = \frac{\ln(x + 2)}{\sqrt{9 - x^2}}$.

a. $(-2, 3)$ ✓

b. $[-2, 3]$

c. $(-3, 3)$

d. $(0, 3)$

$x + 2 > 0 \Rightarrow x > -2$. $9 - x^2 > 0 \Rightarrow -3 < x < 3$. Intersection gives $x \in (-2, 3)$.

Section 4: Trigonometric Identities and Equations

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10 Simplify $\frac{\sin(2\theta)}{1+\cos(2\theta)}$ into a single trigonometric function.

a. $\tan\theta$ ✓

b. $\cot\theta$

c. $\sec\theta$

d. $\sin\theta$

$\frac{2\sin\theta\cos\theta}{1+(2\cos^2\theta-1)} = \frac{2\sin\theta\cos\theta}{2\cos^2\theta} = \frac{\sin\theta}{\cos\theta} = \tan\theta$.

11 Solve $2\cos^2 x + \cos x - 1 = 0$ on the interval $[0, 2\pi)$.

a. $\pi/3, \pi, 5\pi/3$ ✓

b. $\pi/6, \pi, 11\pi/6$

c. $\pi/3, 2\pi/3$

d. $\pi/2, 3\pi/2$

$(2\cos x - 1)(\cos x + 1) = 0 \Rightarrow \cos x = 1/2$ or $\cos x = -1$. On $[0, 2\pi)$, $x = \pi/3, 5\pi/3$, or π.

12 If $\sin\alpha = 3/5$ with $\alpha \in (\pi/2, \pi)$, find $\cos(2\alpha)$.

a. $7/25$ ✓

b. $-7/25$

c. $24/25$

d. $-24/25$

$\cos(2\alpha) = 1 - 2\sin^2\alpha = 1 - 2(3/5)^2 = 1 - 2(9/25) = 1 - 18/25 = 7/25$.

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